Literature Review

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1 Introduction

After discussion with A/Prof Douglas R. Brumley, My literature review will be around ciliary and flagellar flow.

2 Ciliary and flagellar flow

2.1 Definitions

Definition 2.1 (Cilium (pl. Cilia))**.** *Cilia* are short, hair-like protrusions that grow from the surface as an extension of the membrane. [1]

Definition 2.2 (Flagellum (pl. Flagellum))**.** *Flagella* are whip-like appendages that are found in many cells and organisms such as bacteria, sperm cells and algae. [2]

Beating - the periodic, rotating motion of cilia and flagella - is a fundamental behaviour which has been extensively studied since at least (insert year). They serve important functions in motility, sensory reception, signalling, and mass transport. Studies of ciliary and flagellar flows therefore have significant biological implications. [1]

2.2 Corals

Cilia-induced vortical flows are critical for O_2 regulation and metabolite exchange across coral-water interfaces. **Pacherres et al.** [3]

- Methodology: "high-speed imaging with a microscope chamber to determine cilia beating frequencies (CBFs) and particle image velocimetry (PIV) in combination with chemical imaging (SensPIV) for mapping flow and O2 concentration fields of corals exposed to increasing seawater temperature."
- Found that ciliary beating is important for thermal tolerance for corals. By beating with higher frequency, the corals increase O_2 intake, which alleviates hypoxia observed in temperatures $\sim 35\text{C}^\circ$ and above. Above $\sim 37\text{C}^\circ$, coral morality accelerated, indicating the existence of a physiological tipping-point for reef-building corals.

A shortcoming of this study and its conclusions is that it is unclear what specific temperature-dependent dynamic viscosity function was used in the Navier-Stokes equation for the mathematical simulations. Furthermore, as highlighted in a study of mussels by Riisgård and Larsen [4], higher temperatures lead to low viscosity, which in term lead to higher cilia beating frequency.

2.3 Synchronisation

Each cilia and flagella beat with a characteristic rhythm, but studies have shown that in the presence of nearby beating cilia/flagella, the beating can synchronise.[2] The nature of the coordination is often *metachronal* - that is, sequencial and staggered (different phases), creating the appearance of a travelling wave. A macroscopic example of this is the mexican wave. [5]

References

[1] Naoko Mizuno et al. "Structural Studies of Ciliary Components". In: *Journal of Molecular Biology* 422.2 (Sept. 2012), pp. 163–180. ISSN: 0022-2836. DOI: `10.1016/j.jmb.2012.05.040`.

[2] Douglas R Brumley et al. "Flagellar Synchronization through Direct Hydrodynamic Interactions". In: *eLife* 3 (July 2014). Ed. by W James Nelson, e02750. ISSN: 2050-084X. DOI: `10.7554/eLife.02750`.

[3] Cesar O. Pacherres et al. "Acute Temperature Effects on Cilia Beating Increase Coral Deoxygenation". In: *Science Advances* 12.21 (), eaeg0950. DOI: `10.1126/sciadv.aeg0950`. (Visited on 09/02/2026).

[4] Hu Riisgård and Ps Larsen. "Viscosity of Seawater Controls Beat Frequency of Water-Pumping Cilia and Filtration Rate of Mussels Mytilus Edulis". In: *Marine Ecology Progress Series* 343 (Aug. 2007), pp. 141–150. ISSN: 0171-8630, 1616-1599. DOI: `10.3354/meps06930`. (Visited on 08/28/2026).

[5] Douglas R. Brumley et al. "Metachronal Waves in the Flagellar Beating of *Volvox* and Their Hydrodynamic Origin". In: *Journal of The Royal Society Interface* 12.108 (July 2015), p. 20141358. ISSN: 1742-5689, 1742-5662. DOI: `10.1098/rsif.2014.1358`. (Visited on 09/03/2026).